

Consumption of Hibiscus sabdariffa L. aqueous extract and its impact on systemic antioxidant potential in healthy subjects.

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BACKGROUND: To evaluate health benefits attributed to Hibiscus sabdariffa L. a randomized, open-label, two-way crossover study was undertaken to compare the impact of an aqueous H. sabdariffa L. extract (HSE) on the systemic antioxidant potential (AOP; assayed by ferric reducing antioxidant power (FRAP)) with a reference treatment (water) in eight healthy volunteers. The biokinetic variables were the areas under the curve (AUC) of plasma FRAP, ascorbic acid and urate that are above the pre-dose concentration, and the amounts excreted into urine within 24 h (Ae(0-24)) of antioxidants as assayed by FRAP, ascorbic acid, uric acid, malondialdehyde (biomarker for oxidative stress), and hippuric acid (metabolite and potential biomarker for total polyphenol intake). RESULTS: HSE caused significantly higher plasma AUC of FRAP, an increase in Ae(0-24) of FRAP, ascorbic acid and hippuric acid, whereas malondialdehyde excretion was reduced. Furthermore, the main hibiscus anthocyanins as well as one glucuronide conjugate could be quantified in the volunteers' urine (0.02% of the administered dose). CONCLUSION: The aqueous HSE investigated in this study enhanced the systemic AOP and reduced the oxidative stress in humans. Furthermore, the increased urinary hippuric acid excretion after HSE consumption indicates a high biotransformation of the ingested HSE polyphenols, most likely caused by the colonic microbiota. Copyright © 2012 Society of Chemical Industry.